Week 4 – Workshop
Gravity: Density, Structures, and Non-Uniqueness

Adelaide University

Geophysics 3A: Potential Fields & Geothermics

July 11, 2026

Duration: 3 hours **Setting:** Workshop room (whiteboards; projector)

Preparation

Pre-reading: Course Notes, Ch. 6
Lecture videos:
Notes:

Purpose

This workshop develops intuition about how density varies in porous and compacted materials, how different subsurface structures relate to density contrasts, and how the buried sphere solution illustrates non-uniqueness in gravity. Students work from physical reasoning, analytic solutions, and qualitative cross-section analysis. The session prepares students for Week 5 (filters, upward/downward continuation, reduction-to-the-pole).

Learning Outcomes

- Estimate bulk density of porous materials using simple mixing laws.
- Explain how compaction and pore-fluid substitution change density.
- Predict density contrasts for various geologic structures.
- Interpret gravity anomaly width–depth relationships for the buried sphere.
- Recognise non-uniqueness and explain why lenticular shallow bodies can mimic deep spheres.

Session Flow (170 min)

Mixed groups of 4–5 with geology and physics/engineering backgrounds.

0–15 min: Warm-up — What does gravity see?

In groups, rank a series of geological changes according to the density contrasts they are likely to produce. Consider examples such as felsic versus mafic rocks, porous versus compacted sediments, and common metamorphic transformations. Discuss which changes would be most visible to a gravity survey and which might produce similar gravity responses despite having very different geological origins.

Key idea: Gravity does not measure rock names or geologic processes directly—it responds to density contrasts. Different geological processes can therefore produce similar gravity anomalies, introducing non-uniqueness into interpretation.

15–30 min: Mini-lecture

30–70 min: Activity 4.A: Density and Porosity

Examine how porosity evolves during burial and how compaction changes bulk density. Using real porosity–depth data, develop a simple compaction model and explore how the resulting density changes influence gravity interpretation.

70–110 min: Activity 4.B: Geologic Structures to Gravity Anomalies

In this activity, you will discuss in groups some basic structures and how they connect to geology and density contrast. You will then make an educated guess what the gravity anomaly in response to the structure will be.

110–150 min: Activity 4.C: Buried Sphere — Width to Depth

Work with the buried sphere analytic solution to explore non-uniqueness.

Explain why increasing depth increases width but decreasing $\Delta\rho$ does not. Discuss non-uniqueness: why can shallow, lens-shaped bodies ("lenticular") mimic deep spheres?

150–160 min: Wrap-up

1. Which factor (porosity, fluid, composition) most strongly affects density in shallow basins?
2. Which structural density contrast is hardest to distinguish with gravity alone?

160–170 min: Preview: Isostasy

This week we explored how density contrasts produce gravity anomalies. A natural next question is whether large topographic features such as mountain belts should therefore always produce large positive gravity anomalies. Surprisingly, they often do not. Next week we will investigate isostasy—the idea that columns of rock tend toward gravitational balance. We will compare the structure of oceanic and continental lithosphere, examine why topography requires compensating mass variations at depth, and derive the isostatic gravity correction. Along the way, we will see why crustal thickness, mantle density, and lithospheric structure are all important for understanding Earth's large-scale gravity field. Isostasy also provides our first look at how gravity data can constrain deep Earth structure without direct sampling.

Workshop Notes

Workshop Notes

Activity 4.A
Density and Porosity

Purpose

To develop intuition for how porosity affects rock density and how compaction during burial modifies density contrasts relevant to gravity interpretation. By examining real porosity–depth data, students will identify simple mathematical descriptions of compaction and explore how changes in porosity influence the bulk density of sedimentary rocks.

Learning Outcomes

By the end of this activity, you should be able to:

- describe how porosity varies with burial depth and explain the role of compaction,
- identify a plausible mathematical model for porosity reduction with depth and interpret its parameters,
- derive and apply a bulk-density relationship for porous rocks,
- explain how compaction alters density contrasts between sedimentary rocks and crystalline basement,
- relate density changes caused by porosity loss to qualitative gravity responses.

Introduction

In week 2, we calculated several physical properties, though in each case we assumed porosity was negligible. However for many rocks, especially sedimentary rocks near the surface, porosity can be significant.

What is porosity? Porosity is void space, but that doesn't mean it is empty. Other than an unsaturated zone at the surface the voids are generally saturated with fluid (water, brine, steam, oil, gas, H_2). The fraction in sediments can be 30% for porous sandstone up to ~70% at the seafloor. Though for most sedimentary rocks it is <20% and crystalline—plutonic and metamorphic rocks—porosity is often $\lesssim$2%. Porosity comes in a few flavors:

intergranular – between mineral grains, common in sedimentary and volcanic rocks,
vesicular – pore space created by gas bubbles, common in volcanic rocks, particularly shallow erupted basalts,
fracture – cracks and joints created by tectonic activity, decompression near the surface, cooling, or meteorite impacts (minor),
weathering related – generally dissolution by groundwater.

Most porosity loss occurs within the upper ~10–11 km of the crust, where increasing overburden stress closes fractures and compacts sediments. Small amounts of porosity may persist to greater depths.

Porosity is generally reported as a decimal fraction between 0 and 1, or as a percentage of the total volume of the rock. Only porosity will be reported, not the solid component (matrix), so what fraction makes up the this portion?—well the rest of it of course!

Tasks

In the figure below you will see sediment porosity–depth data for a chalk section off the coast of Norway. Estimate the nature of the compaction.

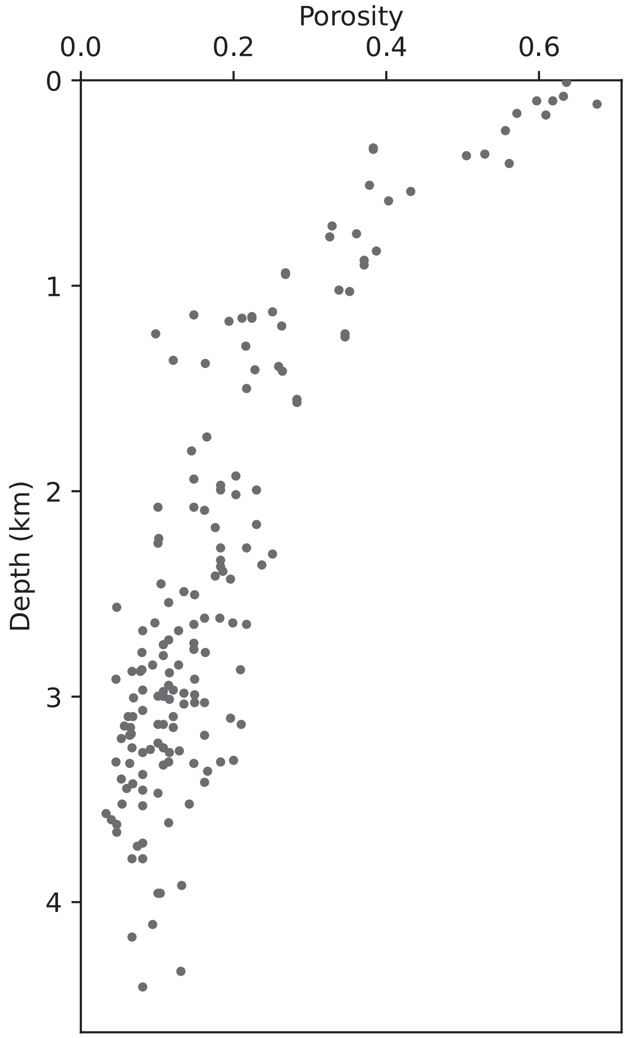


Figure 1: Porosity data from Mallon and Swarbrick (2002) on calcareous ooze.

Question 1. Compaction of Sediments

To estimate the compaction relatioship with depth, start by drawing an approximate curve for porosity from the surface to 4 km depth.

Which of the mathematical models appears to fit the data best? And, why do all of these models include a z/h term?

a. $\phi(z) = \phi_0 - zh^{-1}$ b. $\phi(z) = \phi_0 \exp\left(-zh^{-1}\right)$ c. $\phi(z) = \dfrac{\phi_0}{1 + zh^{-1}}$

Assuming your preferred model, estimate values for ϕ_0 and h? Also label these on the figure on the appropriate axes.

What physical meaning would you assign to: (a) ϕ_0, (b) h?

Question 2. Density of Sediments

If the density of the matrix material, ρ_m, in the chalk examined above ranges from 2650 to 2720 kg m^{-3}, derive a relationship for the bulk density of the sediment with pores filled with seawater ($\rho_f = 1030$ kg m^{-3}).

Which depth interval produces the largest change in density: 0–1 km, 1–2 km, 2–3 km, or 3–4 km? Why?

What is the density of the sediments at the surface and h? What is the relative change in density?

Question 3. From Porosity to Gravity

As sediments become buried and compacted, do you expect the density contrast between the sediments and crystalline basement to increase or decrease?

What implications does this have for gravity interpretation?

Key Ideas

Porosity and density are linked. Sedimentary rocks contain pore space that is commonly filled with fluids. As porosity decreases, the fraction of solid material increases and the bulk density moves closer to the density of the rock matrix.

Compaction is most effective at shallow depths. Increasing overburden stress reduces pore space, but the rate of compaction decreases with depth as sediments become progressively denser and less compressible. Most porosity loss occurs in the upper crust.

Density contrasts evolve during burial. Because compaction changes the density of sediments, the density contrast between sediments and crystalline basement changes through time. These changes influence the gravity field and affect how sedimentary basins are interpreted.

Activity 4.B
Geologic Structures to Gravity Anomalies

Purpose

To develop intuition for how geological structures are represented by idealized geometric bodies in gravity interpretation. By relating common geologic features to simple analytical models, you will explore how density contrast, geometry, and burial depth influence the character of gravity anomalies.

Learning Outcomes

By the end of this activity, you should be able to:

- identify geological structures that can be approximated by common gravity-model geometries,
- estimate plausible density contrasts for a variety of lithologies and geologic settings,
- predict the qualitative shape of gravity anomalies from simple subsurface structures,
- relate anomaly symmetry, width, amplitude, and gradients to source geometry,
- explain the advantages and limitations of using simplified geometric models in gravity interpretation.

Introduction

The structural arrangement of rock units can be very complex. As geophysicists we invert for more complex structures as necessary, but doing so is time-consuming. We can often approximate complex structures by one or more simplified geometries. These simplifications have benefits besides being faster. For one, the models have closed form analytical solutions. We can calculate a wide range of stuctures that vary in dimension, distance, or orientation rapidly giving us approximate solutions that in many cases are 'good enough' given all the data uncertainties and model non-uniqueness issues. Another advantage is that closed form solutions help build our intution in a way that inverting for cell properties do not.

Task

For each structure below, complete the table and use the *upper axes* to sketch the expected gravity anomaly $g_z(x)$ (arbitrary units). The *lower panel* shows a schematic cross-section (z positive downward). Assume that structures extend into and out of the page (i.e., perpendicular to strike).

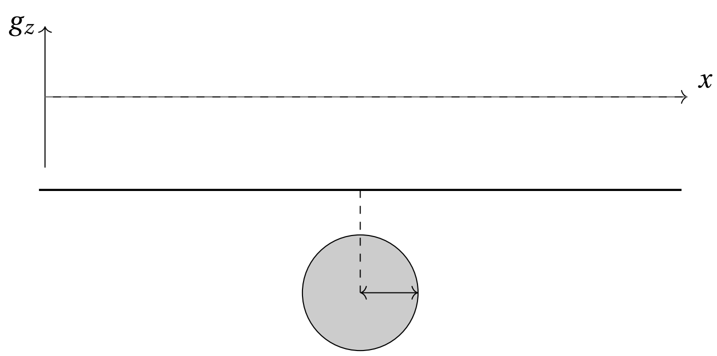
In your groups:

1. Discuss geologic structures that may be described by the simple shapes.
2. Propose plausible density contrasts ($\Delta\rho$) given typical lithologies.
3. Predict qualitative anomaly shape: narrow/broad? symmetric/asymmetric? positive/negative? asymptotic limit?
4. One group presents each structure; whole class discusses similarities and differences.

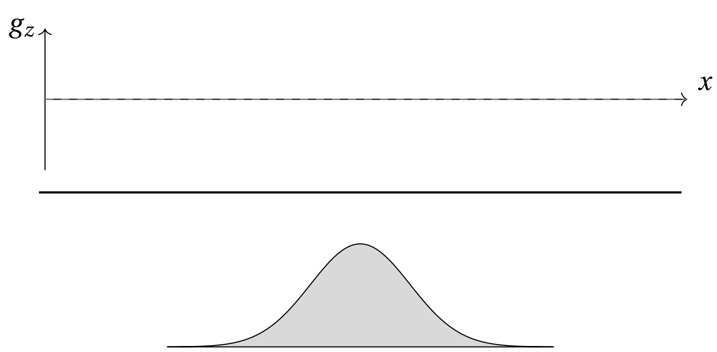
	Geologic Structure(s)	Likely lithologies	Plausible Range $\Delta\rho$ (kg m^{-3})
A.			
B.			
C.			
D.			
E.			
F.			
G.			

Prompts: Indicate sign (pos./neg.), width (narrow/broad), symmetry/asymmetry, and where the steepest gradients occur.

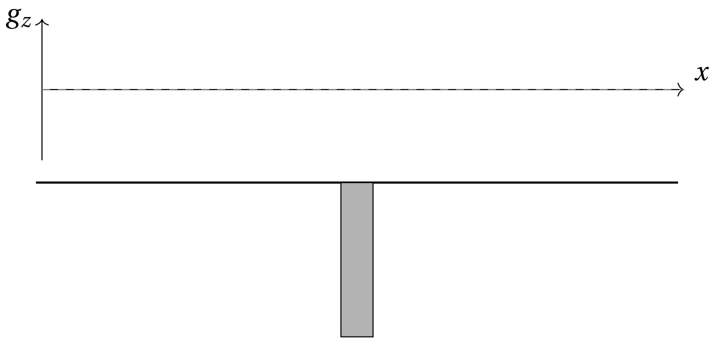
A) Cylinder



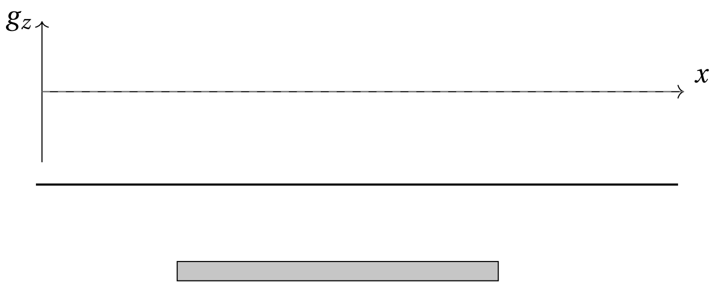
B) Sigmoid



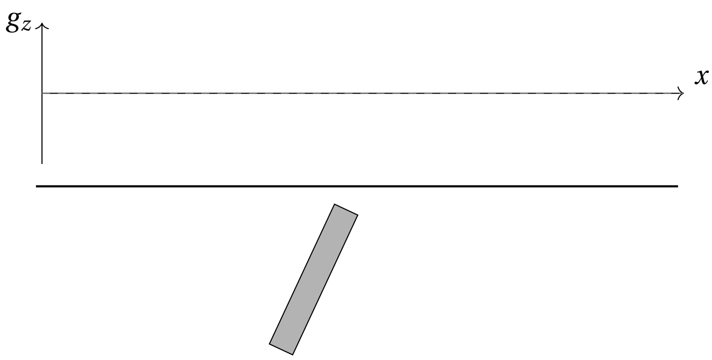
C) Vertical Slab



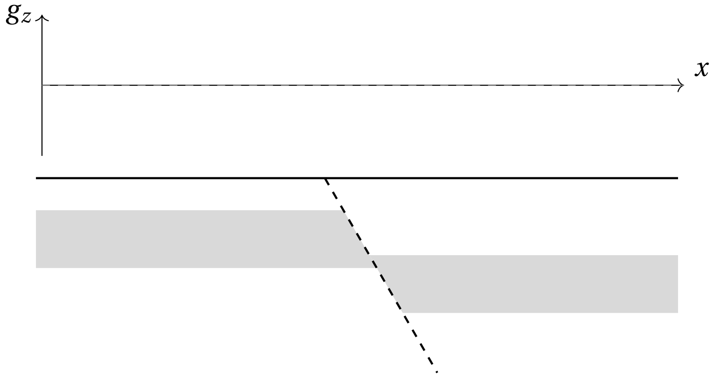
D) Finite-Extent Horizontal Slab



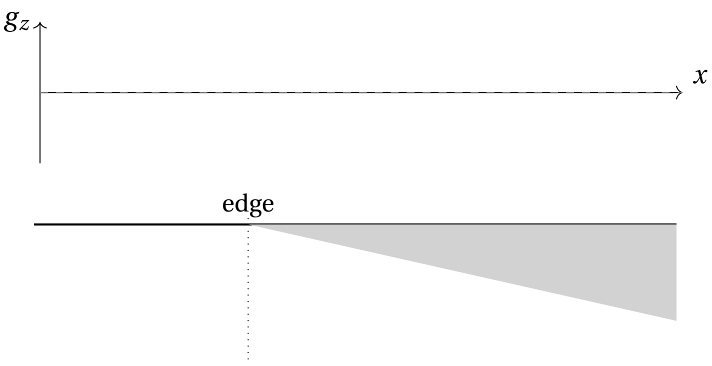
E) Inclined Plane



F) Offset Layer



G) Gently Sloped Basin (edge at one-third from left boundary)



Key Ideas

Anomaly width reflects source depth. Shallower bodies produce narrower, sharper anomalies; deeper bodies produce broader, smoother anomalies. This depth–wavelength relationship is a fundamental constraint on gravity interpretation.

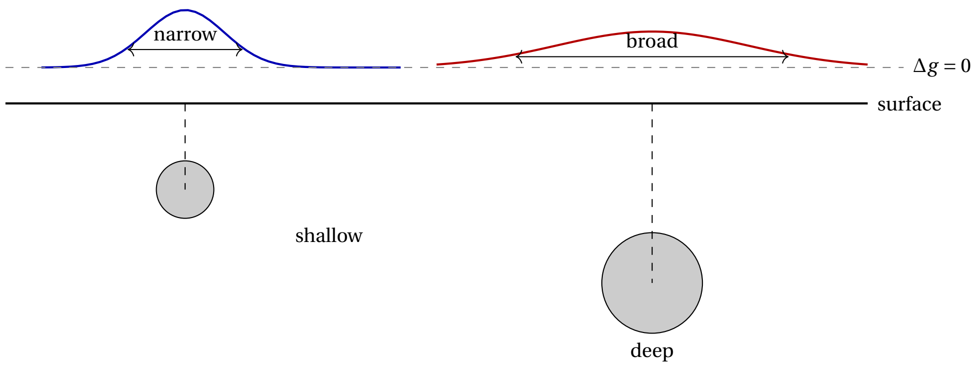


Figure 2: Shallower bodies produce narrower, higher-amplitude anomalies; deeper bodies produce broader, lower-amplitude anomalies with the same total mass excess. Anomaly half-width ≈ depth to source.

Anomaly sign reflects density contrast. Bodies denser than their surroundings ($\Delta\rho > 0$) produce positive anomalies; bodies less dense than their surroundings produce negative anomalies.

Symmetry reveals geometry. Symmetric anomalies suggest symmetric bodies (sphere, vertical dike); asymmetric anomalies indicate asymmetric structures (inclined plane, faulted contact). The location and sign of the steepest gradient constrain the lateral position and dip of the source edge.

Non-uniqueness is unavoidable. Different geometries with appropriate density contrasts can produce identical surface gravity anomalies. Additional constraints (geology, seismic data, drill-hole information) are always needed to discriminate between alternative models.

Activity 4.C
Buried Sphere — Width to Depth

Purpose

This activity derives and applies a simple depth-estimation rule for gravity anomalies using the buried sphere model. By working through the mathematics of the full-width at half-maximum, you will connect the observable width of an anomaly to the depth of its source, independent of the source mass — a result that illustrates both the utility and the non-uniqueness of gravity interpretation.

Learning Outcomes

By the end of this activity, you should be able to:

- Relate observable characteristics of a gravity anomaly to source geometry and depth.
- Derive a simple relationship between anomaly width and source depth from the buried sphere model.
- Explain why some source parameters influence anomaly amplitude whereas others influence anomaly shape.
- Describe sources of non-uniqueness in gravity interpretation and suggest ways to reduce them.
- Use simple analytical results to guide preliminary geological interpretation prior to inversion.

Introduction

The gravity anomaly due to a buried sphere is given by:

$$g_z(x) = \tfrac{4}{3}\pi a^3 G \Delta\rho \frac{z_c}{(x^2 + z_c^2)^{3/2}},$$

where a is the radius of the sphere, Δrho is its density anomaly, and z_c is the depth to the center of the sphere. The coordinate x is the position along the profile at Earth's surface.

Last week, we saw this relationship leads to non-uniqueness because $a^3\Delta\rho$ are inseperable. Now, however, we want to show that one can produce a simple relationship between the depth of the buried sphere and the width of its anomaly, independent of the the mass. Such a relationship is useful for estimating the depth to structures that produce peaks in the gravity field. Such a relationship is useful for producing back of the envelope estimates in the field or from a map before any formal modeling is done. Such relationships are the prelude concepts that lead to wavelength filtering that we will see in Week 7.

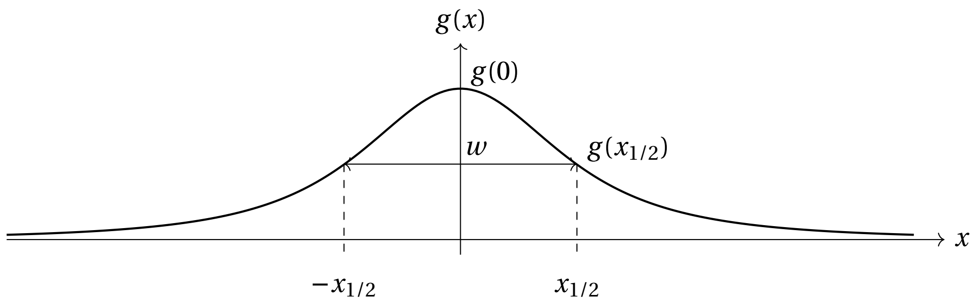


Figure 3: Gravity anomaly due to a buried sphere.

Tasks

To develop the depth to width relationship:

Question 1. Peak Anomaly
Show that $g_z(0)$ is proportional to $\Delta\rho\, a^3/z_c^2$.

Question 2. Width to Depth Relationship
Show that the depth

$$z_c \approx 0.65\, w$$

where w is the full-width half-maximum.

To do so, solve for $x_{1/2}$ starting from $g_z(x_{1/2}) = \frac{1}{2} g_z(0)$. Hint: this is on the right track,

$$\frac{z_c}{x_{1/2}} = (2^{2/3} - 1)^{-1/2}.$$

Question 3. Interpretation

a Why does depth affect width but density contrast does not?

36pt

b How do the radius and density contrast result create non-uniqueness for inversion and how we ensure uniqueness?

36pt

c Why does a shallow lenticular body (e.g., laccolith or lopolith) could mimic a deep sphere's anomaly?

36pt

Question 4. When Depth-Wavelength Breaks Down

Explain why sedimentary basins can give broad anomalies while being shallow features.

Postulate how sedimentary basins can then be distinguished from deep features.

Key Ideas

Different source parameters influence different characteristics of an anomaly. Changes in density contrast primarily affect anomaly amplitude, changes in source geometry affect anomaly shape, and changes in depth affect anomaly width (or wavelength).

Not all source properties are equally well constrained. Several combinations of geometry and density contrast can produce similar gravity anomalies. Understanding which parts of an anomaly are controlled by which parameters helps identify sources of non-uniqueness.

Depth and wavelength are fundamentally related. Shallow sources tend to produce narrow, short-wavelength anomalies, whereas deep sources produce broader, long-wavelength anomalies. This relationship underpins many depth-estimation and filtering techniques used in potential-field geophysics.